

3.12.3 Special relativity (A-level only)

3.12.3.1 The Michelson–Morley experiment (A-level only)

Content

Principle of the Michelson–Morley interferometer.

Outline of the experiment as a means of detecting absolute motion.

Significance of the failure to detect absolute motion.

The invariance of the speed of light.

3.12.3.2 Einstein's theory of special relativity (A-level only)

Content

The concept of an inertial frame of reference.

The two postulates of Einstein's theory of special relativity:

1. physical laws have the same form in all inertial frames
2. the speed of light in free space is invariant.

3.12.3.3 Time dilation (A-level only)

Content

Proper time and time dilation as a consequence of special relativity.

Time dilation:

$$t = \frac{t_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Evidence for time dilation from muon decay.

3.12.3.4 Length contraction (A-level only)

Content

Length of an object having a speed v

$$l = l_0 \sqrt{1 - \frac{v^2}{c^2}}$$

Equivalence of mass and energy, $E = mc^2$; $E = \frac{m_0 c^2}{\sqrt{1 - \frac{v^2}{c^2}}}$

Graphs of variation of mass and kinetic energy with speed.

Bertozzi's experiment as direct evidence for the variation of kinetic energy with speed.